Data Model and Local Storage

1. Principles

Local-first storage is assumed by the PRD. Verify the actual schema before introducing changes. Minimize personal data; use migrations; keep secrets outside ordinary tables; handle associated files consistently during deletion.

2. Proposed entities

customers

`id`, `name`, `company_name?`, `phone?`, `email?`, `notes?`, `created_at`, `updated_at`.

insured_members

`id`, `quote_id`, `name`, `relationship`, `age?`, `date_of_birth?` only if necessary, insurer-required fields, timestamps. Avoid storing both age and DOB without a documented reason.

quotes

`id`, `customer_id`, `product_type`, `sum_insured`, `cover_type` (floater/individual), `pincode`, `status`, timestamps, retention metadata.

insurer_results

`id`, `quote_id`, `run_id`, `insurer_id`, `status`, `premium_value`, `currency`, `source_type`, `plan_label`, `coverage_summary`, `retrieved_at`, `error_code`, `safe_message`, `evidence_reference`.

quote_events

`id`, `quote_id`, `run_id?`, `event_type`, `occurred_at`, sanitized details. Do not store secrets or full member payloads in event details.

settings/branding

Non-secret settings, approved logo reference, business name/address/contact, approved licence/GST fields, disclaimer/consent text, retention period.

3. Monetary data

Use integer paise or another documented decimal-safe canonical representation. Do not use binary floating-point as the canonical value. Format ■ values only at display/PDF layer. Never rank null, invalid or ambiguous amounts.

4. Credential storage

Use an OS-backed secure store. SQLite may hold a non-secret key/reference but not raw passwords, OTPs, tokens or session cookies. If secure storage is unavailable, fail safely and do not fall back to plaintext.

5. Integrity and migrations

Version migrations; enable foreign keys where supported; transact related writes; index common quote/customer/date lookups; test upgrade from previous release; define recovery behavior for locked/corrupt DB.

6. Retention/deletion

Owner must approve N-day policy. Expiry should cover database rows, linked screenshots and app-managed temporary artifacts. Clearly state that exported files saved elsewhere may remain outside app control. Test deletion and record a sanitized audit event.

7. Backup and recovery

Define whether local backups are supported, how they are protected, how restore works, how migration failures are handled, and how users are warned before reset/reinstall.